

RESHORE 2040 Pharmaceutical Biomanufacturing Initiative:

Scalable Biomanufacturing Platforms for Essential Medicines

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Vision Statement

Despite having the third-largest pharmaceutical industry in the world, Japan still imports roughly 60-70% of essential medicines from abroad due to the unsustainable economics of domestic production.¹ This dependence on foreign suppliers results in drug shortages that affect more than 20% of Japan's essential medicines, posing an existential risk to public health and national security.^{2,3} The RESHORE 2040 Pharmaceutical Biomanufacturing Initiative (RESHORE 2040) is an ARPA styled research program that seeks to eliminate drug shortages resulting from pharmaceutical supply chain disruptions. This program will revolutionize Japan's pharmaceutical manufacturing capacity by developing and de-risking disruptive new technology platforms that enable the domestic production of essential medicines at economically sustainable price points, reducing the frequency of drug shortages by as much as 95%.

H1. What are you trying to do?

The RESHORE 2040 program will eliminate drug shortages through three specific aims that emphasize simplicity, efficiency, and cost-competitiveness. First, this program will leverage biomanufacturing to consolidate the pharmaceutical supply chain around domestically available feedstocks. Second, this program will develop advanced manufacturing platforms to efficiently switch between drug products and scales of production. Third, this program will demonstrate the cost-competitive production of essential medicines, compared to those sourced from the People's Republic of China. Success in these three aims would reduce the frequency of drug shortages by 95%, reduce shortage-associated deaths and hospitalizations by 90%, and set Japan on a course to achieve pharmaceutical independence by as early as 2040.⁴

H2. How is it done today, and what are the limits of current practice?

Today, batch manufacturing represents the industry standard in pharmaceutical drug production. Traditional batch manufacturing facilities are exceptionally large, possessing footprints on the order of ~10,000-100,000m², making them incredibly expensive to construct, as well as maintain.^{5,6} Because of their large size, batch manufacturing facilities suffer from low equipment utilization rates (between 50-60%) and long changeover times (in excess of 2 weeks).^{7,8} Because they are designed to produce drugs at a fixed scale, these facilities do not possess the ability to surge production in the event of drug shortages. Taken together, these constraints emphasize the need for more flexible, more responsive approaches to the production of essential medicines.

H3. What is new in your approach and why do you think it will be successful?

The Bio Strategy 2019 (バイオ戦略) has motivated significant investments into Japan's bioeconomy, including the biotechnology, biomanufacturing, and life sciences sectors.⁹ The proposed ARPA styled program is novel in that it seeks to leverage these newly developed biotechnology sectors to support and sustain the development of the disruptive new biomanufacturing platforms proposed herein. By choosing to focus on the most heavily regulated and capital-intensive sector first (*i.e.*, the pharmaceutical industry), this program will be successful in seeding new technologies and startups that can disrupt not only the pharmaceutical industry, but many other, potentially more accessible markets within Japan as well.

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This program will solicit proposals from interdisciplinary teams of researchers from Japanese academia, national laboratories, and the industrial manufacturing sectors. Prospective teams will be asked to propose the development of new technology platforms for the domestic production of essential medicines. While this program will be “solution agnostic,” several compelling and innovative technology approaches are especially well positioned to meet the articulated success metrics for this program. Biomanufacturing solutions may include, but are not limited to: 1) continuous manufacturing and 2) modular process design.

Continuous manufacturing innovations stand to advance manufacturing flexibility and responsiveness by 1) reducing the size of manufacturing facilities, 2) maximizing equipment utilization rates, and 3) scaling production to meet variable demand.^{10–13} Compared to batch, continuous manufacturing facilities may occupy up to 90% less space (on the order of ~100-1000m²), significantly reducing manufacturing costs.^{10,13} Second, by decreasing the size of the equipment needed to manufacture drugs, equipment utilization rates can be made to exceed 90%, increasing process efficiency.¹² Third, by extending process run time, continuous manufacturing can be surged in times of drug shortages, offering a more responsive approach to pharmaceutical supply chain security than can be achieved with the current state-of-the-art.

Separately, modular process designs also stand to advance the flexibility and responsiveness of pharmaceutical manufacturing through the packaging of complex manufacturing units into mobile, reconfigurable “skids.”¹⁴ These skids can be rapidly deconstructed, re-assembled, and deployed for the production of different drugs and therapeutics, pushing changeover times close to their theoretical minimum of 3 days.¹² Whereas conventional manufacturing equipment is fixed in place, modular process designs create interchangeable “process blocks” through the standardization of equipment interfaces, utility connections, and control systems.¹⁵ Today, process changeovers may take several weeks to complete, but modularization could enable “snap” changeovers within hours, setting a new standard for pharmaceutical manufacturing efficiency.

H4. What is the impact if successful? If you are successful, what difference will it make?

If successful, this program would develop technology platforms that reduce the frequency of drug shortages by 95%, reduce shortage-associated deaths and hospitalizations by 90%, and set Japan on a course to achieve robust pharmaceutical independence by as early as 2040. Securing the pharmaceutical supply chain would have major implications on Japan’s national security, as well as the national security of its global allies. As countries around the world begin to appreciate the extent to which China dominates the pharmaceutical supply chain, this program could position Japan to overtake China as the more trustworthy, more reliable supplier of essential medicines, revitalizing Japan’s domestic pharmaceutical manufacturing ecosystem and reinforcing its image as a global leader in pharmaceutical innovation.

H5. What are the risks?

To succeed, this ambitious research program must address three types of risks related to technical execution, regulatory compliance, and commercialization. 1) Technical risks include failure to meet the aggressive technical performance metrics defined by this program. For instance, performers may struggle to produce drugs of consistently quality during extended, continuous production campaigns. This risk will be mitigated by early proof-of-concept demonstrations, rigorous technology down-selection, and pilot-

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scale validation. 2) Regulatory risks include failure to develop equipment qualification frameworks for the manufacturing platforms developed on this program. For instance, it may prove difficult for regulators to define qualification criteria for entirely new types of manufacturing technologies. This regulatory risk will be mitigated through early and sustained engagements with Pharmaceutical and Medical Devices Agency (PMDA) regulatory experts, as well as alignment with recently established regulatory precedent for continuous tableting technologies. 3) Commercialization risks include failure to transition the technology platform to industry due to hesitations or resistance to embracing unvalidated technology platforms. This risk will be mitigated through extensive technology validation and targeted industry engagement activities. These activities may include industry attendance during annual program reviews, educational manufacturing demonstrations, and targeted publicity emphasizing the revolutionary potential of these advanced manufacturing platforms.

H6. How much will it cost? H7. How long will it take?

The RESHORE 2040 program will cost ¥3.2 billion JPY (\$20 million USD) over the course of 3 years.

H8. What milestones will be used to check for success?

Mid-term success will be evaluated based on the performer's ability to demonstrate the pilot-scale production of one essential drug product with a 50% reduction in manufacturing costs, compared to batch manufacturing at a comparable scale. Final program success will be evaluated based on the performer's ability to demonstrate the industrial-scale production of one essential drug product with a 90% reduction in manufacturing costs, compared to batch manufacturing at a comparable scale. Manufacturing costs will be calculated via technoeconomic analysis (TEA), to include equipment purchase costs, manufacturing footprint, production rate, equipment utilization rate, yield, changeover time, downstream purification costs, labor, and utility costs, among others.

Commercialization & Transition Plan

In order to ensure that the technical products developed under this program are transitioned to industry, this program will work with the Ministry for Economy, Trade, and Industry (METI), the Ministry for Health, Labor, and Welfare (MHLW), and the Pharmaceutical and Medical Devices Agency (PMDA) to develop these technologies in line with regulatory compliance and oversight requirements. From industry, this program will engage heavily with the Japanese Pharmaceutical Manufacturers Association (JPMA) and Japan Generic Medicines Association (JGA) as critical commercialization partners with which to transition this technology to industry. This program will be successful if at least one Japanese pharmaceutical manufacturing company files a "Partial Change Approval" application with the PMDA to incorporate the resulting technology platform into their process within one year of the conclusion of this program.

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RESHORE 2040 – Program Structure					
Phase	Months	TA1: Synthesis	TA2: Manufacturing	TA3: Process Flexibility	Budget (USD) / Budget (JPY)
1	9	Objectives: 1) Demonstrate a 50% reduction in the number of chemical inputs needed to produce 1 essential drug molecule at benchtop-scale. 2) Demonstrate the consolidation of at least 3 chemical reactions within 1 bio-enabled transformation. 3) Demonstrate a 5x reduction in hazardous waste compared to batch manufacturing reference. Deliverables: 1) Bill of Materials (BOM). 2) Representative analytical chemistry data. 3) Hazardous waste data, including waste type, volume, and cost of disposal and/or remediation.	Objectives: 1) Demonstrate a 10x reduction in manufacturing footprint compared to batch manufacturing. 2) Demonstrate a 1.5x improvement in equipment utilization rate. 3) Demonstrate the ability to surge production by 2x (volumetric throughput or duration) while maintaining ≥90% of nominal yield. Deliverables: 1) Process Flow Diagram 2) Process Operating Schedule 3) Yield and purity data for the target drug molecule, byproducts, and impurities over the course of a surge.	Objectives: 1) Demonstrate a 2x reduction in changeover time. 2) Standardize 50% of all equipment interfaces, utility hookups, and control systems. 3) Demonstrate platform maturity through equipment qualification & process validation. Deliverables: 1) "Mini-Skid" prototype. 2) Process Integration Plan w/ technical diagrams. 3) Design Qualification (DQ) report, Operational Qualification (OQ) report, & qualification data packages.	\$6M / ¥1B
Pressure Test #1: Demonstrate continuous, benchtop-scale bioproduction of 1 essential drug molecule over the course of 3 days AND demonstrate platform compatibility for the production of at least two other essential drug molecules from different drug classes.					
2	9	Objectives: 1) Demonstrate a 75% reduction in the number of chemical inputs needed to produce 1 essential drug molecule at pilot-scale. 2) Demonstrate the consolidation of at least 5 chemical reactions within 1 bio-enabled transformation. 3) Demonstrate an 8x reduction in hazardous waste compared to batch manufacturing reference. Deliverables: 1) Bill of Materials (BOM). 2) Representative analytical chemistry data. 3) Hazardous waste data, including waste type, volume, and cost of disposal and/or remediation.	Objectives: 1) Demonstrate a 50x reduction in manufacturing footprint compared to batch manufacturing. 2) Demonstrate a 1.75x improvement in equipment utilization rate. 3) Demonstrate the ability to surge production by 4x (volumetric throughput or duration) while maintaining ≥90% of nominal yield. Deliverables: 1) Process Flow Diagram 2) Process Operating Schedule 3) Yield and purity data for the target drug molecule, byproducts, and impurities over the course of a surge.	Objectives: 1) Demonstrate a 3.5x reduction in changeover time. 2) Standardize 75% of all equipment interfaces, utility hookups, and control systems. 3) Demonstrate platform maturity through equipment qualification & process validation. Deliverables: 1) "Pilot-Skid" prototype 2) Process Integration Plan w/ technical diagrams 3) Performance Qualification (PQ) report & qualification data package	\$6M / ¥1B
Pressure Test #2: Demonstrate the continuous, pilot-scale bioproduction of 2 essential drug molecules over the course of 5 days each AND demonstrate platform compatibility for the production of 2 other essential drug molecules from different drug classes.					
Mid-Term Success Metric: Demonstrate the pilot-scale production of 1 essential drug product with a 50% reduction in manufacturing cost compared to batch manufacturing at a comparable scale.					
3	18	Objectives: 1) Demonstrate a 90% reduction in the number of chemical inputs needed to produce 1 essential drug molecule at industrial-scale. 2) Demonstrate the consolidation of at least 8 chemical reactions within 1 bio-enabled transformation. 3) Demonstrate an 10x reduction in hazardous waste compared to batch manufacturing reference. Deliverables: 1) Bill of Materials (BOM). 2) Representative analytical chemistry data. 3) Hazardous waste data, including waste type, volume, and cost of disposal and/or remediation.	Objectives: 1) Demonstrate a 100x reduction in manufacturing footprint compared to batch manufacturing. 2) Demonstrate a 1.8x improvement in equipment utilization rate. 3) Demonstrate the ability to surge production by 10x (volumetric throughput or duration) while maintaining ≥90% of nominal yield. Deliverables: 1) Process Flow Diagram 2) Process Operating Schedule 3) Yield and purity data for the target drug molecule, byproducts, and impurities over the course of a surge.	Objectives: 1) Demonstrate a 5x reduction in changeover time. 2) Standardize 100% of all equipment interfaces, utility hookups, and control systems. 3) Summarize equipment qualification in a "Partial Change Approval" application to support rapid industrial adoption. Deliverables: 1) "Industrial-Skid" prototype 2) Process Integration Plan w/ technical diagrams 3) Regulatory-ready "Partial Change Approval" application & equipment qualification data packages.	\$8M / ¥1.2B
Pressure Test #3: Demonstrate the continuous, industrial-scale bioproduction of 3 essential drug molecules in a relevant GMP environment for 7 days AND meet or exceed program metrics for each Technical Area (TA).					
Final Success Metric: Demonstrate the industrial-scale production of 1 essential drug product with a 90% reduction in manufacturing cost compared to batch manufacturing at a comparable scale.					

Figure 2. A detailed overview of the RESHORE 2040 program structure broken down by TA, including objectives, deliverables, and phase-gate success criteria.

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